

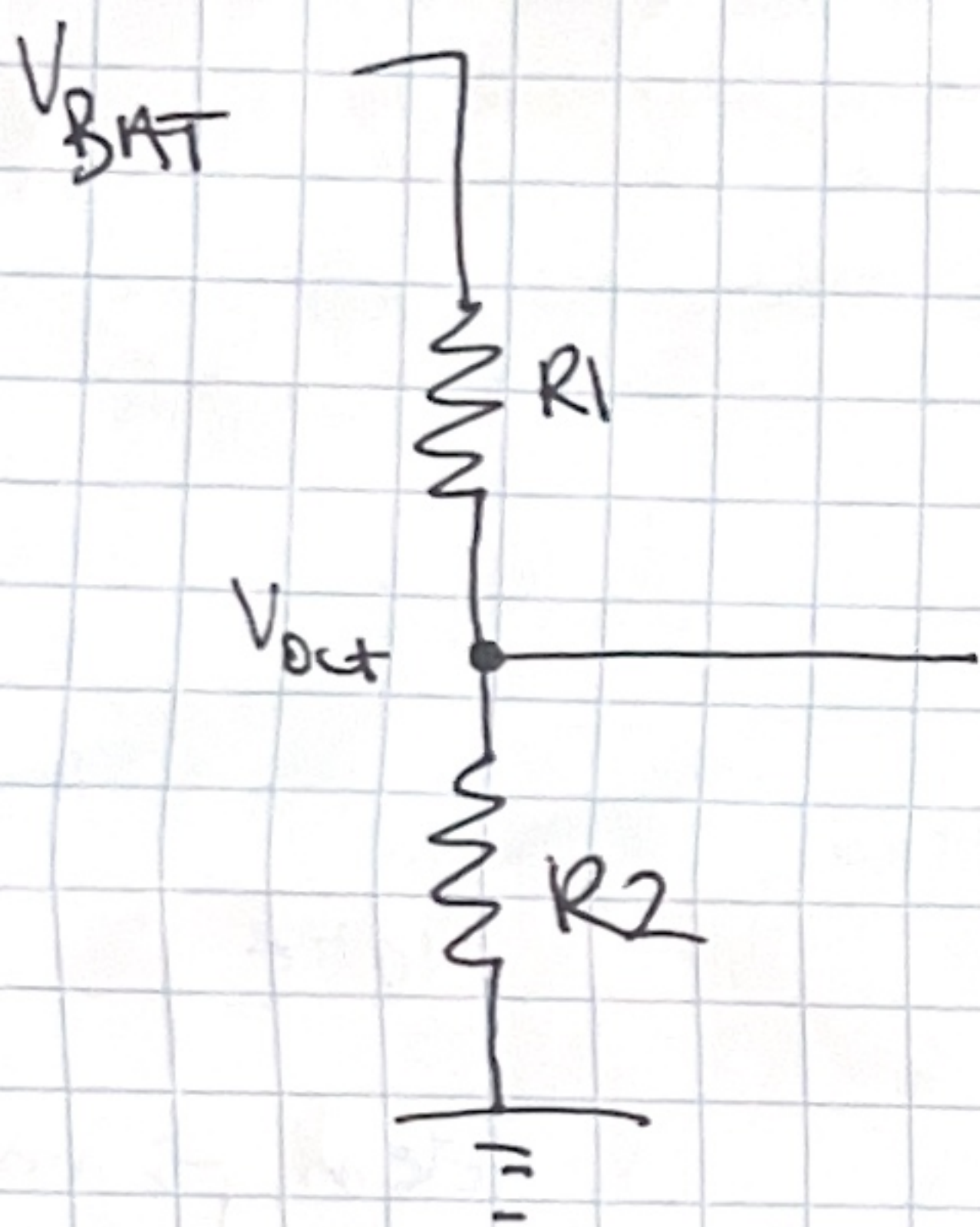
→ V_{ref} Median is 1.1V

→ Battery voltage is 3.7V - 4.2V (Nominal)

↳ To add extra range, go from 0-5V.

→ To Capture 0-5V, 5V needs to become 1.1V to be read by ADC.

→ Want to use components I have. Set $R_1 = 680k$



$$V_{out} = V_{BATT} \left(\frac{R_2}{R_1 + R_2} \right)$$

$$V_{out}(R_1 + R_2) = V_{BATT} R_2$$

$$V_{out} R_1 + V_{out} R_2 = V_{BATT} R_2$$

$$V_{out} R_1 = V_{BATT} R_2 - V_{out} R_2$$

$$\frac{V_{out} R_1}{V_{BATT} - V_{out}} = R_2$$

For $V_{BATT} = 5V$, we want $V_{out} = 1.1V$,
 $R_1 = 680k$

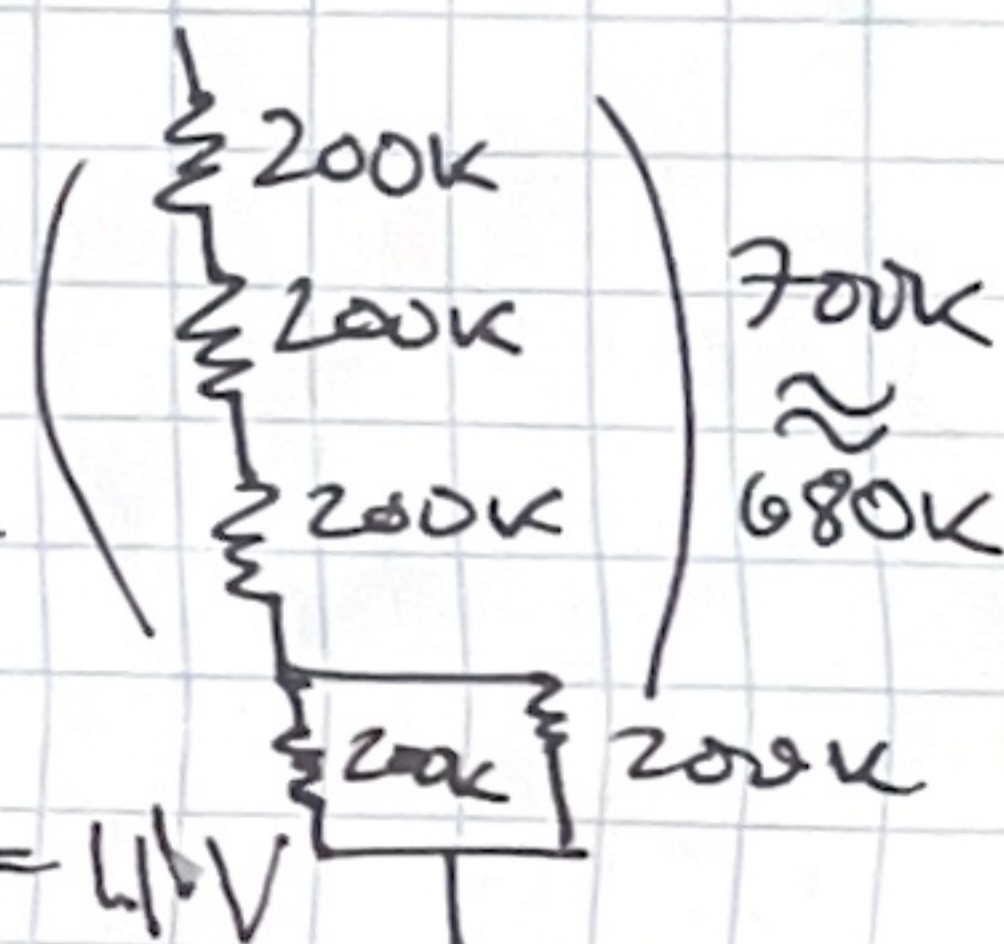
$$R_2 = \frac{(1.1)(680k)}{(5 - 1.1)} = 192k\Omega$$

→ (closest I have is 200k).

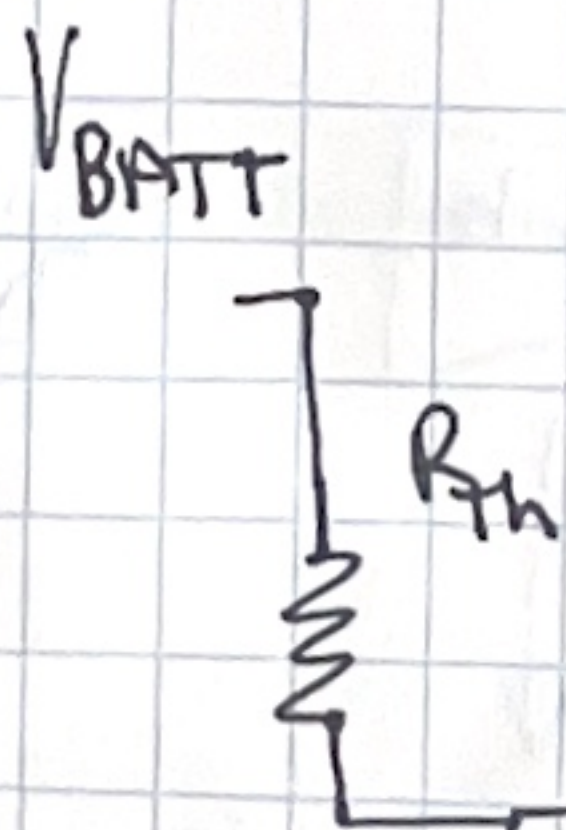
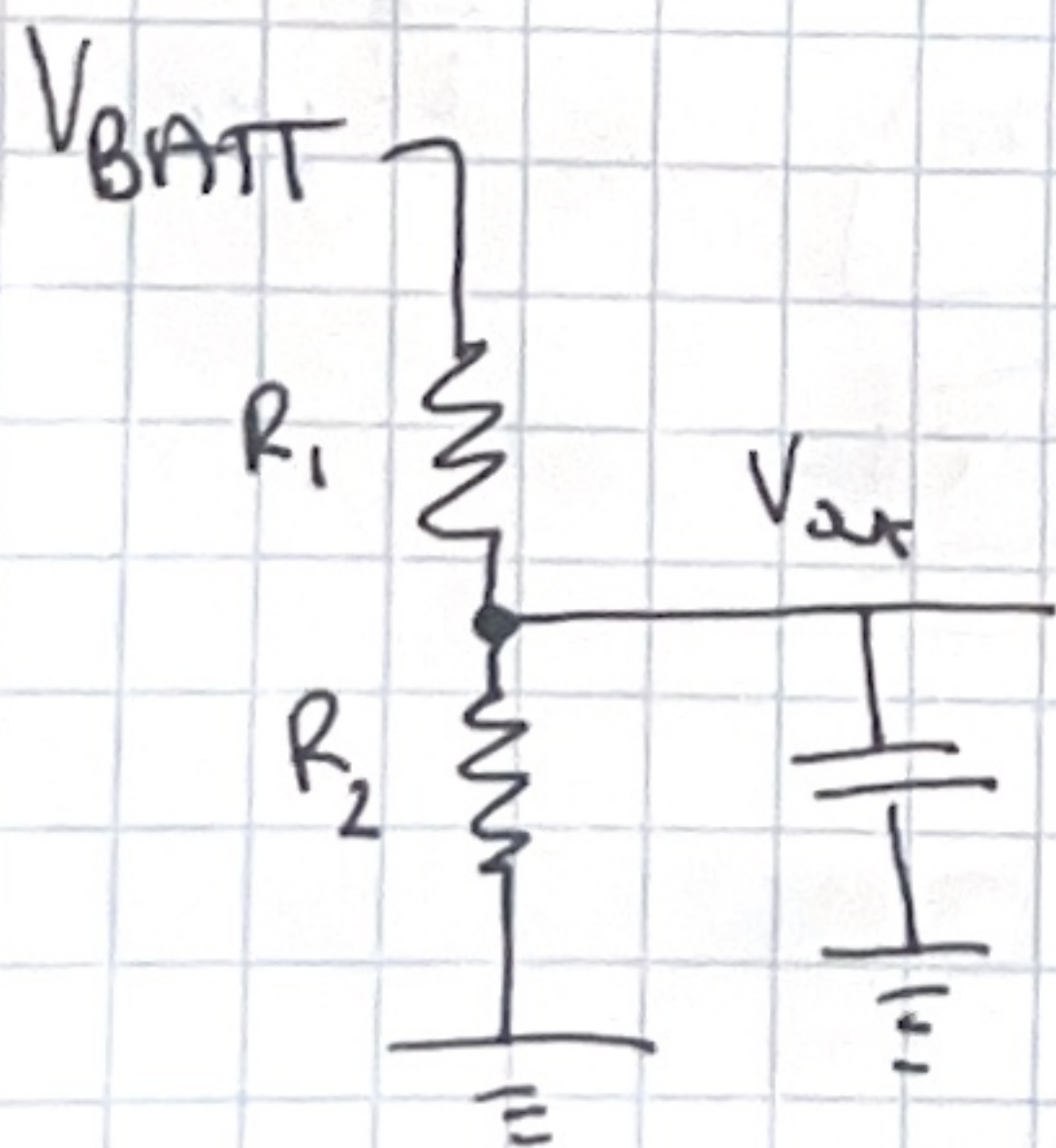
* Note *

I only have 200k resistors actually.

Going for



Filter Design



$$R_{th} = (R_1 \parallel R_2) = (680k \parallel 200k)$$

$$R_{th} = (150k \text{ ohms})$$

Cutoff Frequency:

$$R_{th} R = 150k \quad C = 0.1\mu F$$

$$f_c = \frac{1}{2\pi R C} = 10.61 \text{ Hz}$$

* Sample at 100 Hz for
10x aliasing margin.